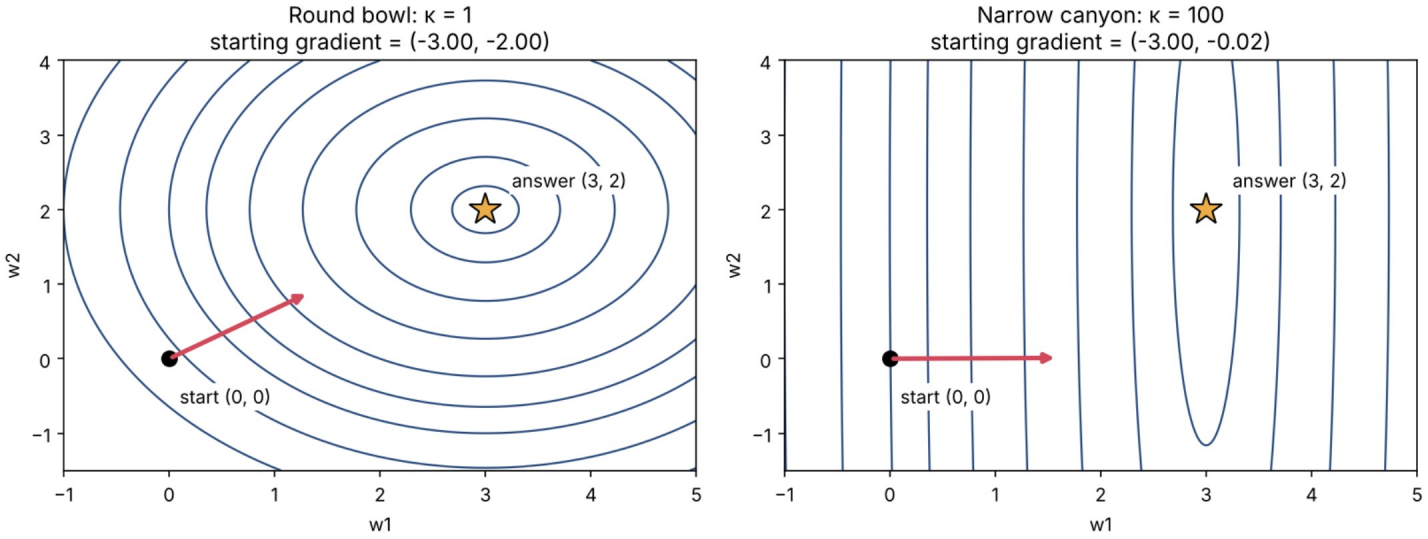
Why training slows down on narrow loss valleys



Red arrow = direction gradient descent first moves

150×

stronger pull along w_1 than w_2 at $\kappa = 100$

only 1.5× on the round bowl ($\kappa = 1$)

DOMAIN

Optimization for machine learning: training a model by minimizing a loss

CHALLENGE

A stretched loss surface (large κ) makes gradient descent zig-zag and need many steps

Group 7 synthetic datasets, κ = 1, 10, 100 · start w = (0, 0) · red arrow = first gradient step